

Question number	Answer	Mark
1(a)	The only correct answer is D <i>A is incorrect because antigens do not protect the body from infection</i> <i>B is incorrect because antigens do not protect the body from infection</i> <i>C is incorrect because interferon does not prevent infection</i>	(1)

Question number	Answer	Additional guidance	Mark
1(b)(i)	An answer that includes the following points: • water alone does not {remove / kill} all microorganisms / soap more effective (at removing microorganisms) (1) • more microorganisms are {removed / killed} as the time of washing increases (with both methods) (1)	ACCEPT skin flora / bacteria cleaner area IGNORE viruses / pathogens ACCEPT skin flora / bacteria cleaner area IGNORE viruses / pathogens	(2)

Question number	Answer	Additional guidance	Mark
1(b)(ii)	An explanation that includes two of the following points: • water alone will {physically remove / wash off} some of the skin flora (1) • soap {contains antimicrobials / contains chemicals / has a high pH} that kill skin flora (1) • the longer the hands are washed in soap the longer the {antimicrobials / chemicals / high pH} have to {affect / kill} skin flora (1) • (some microorganisms left) soap cannot {destroy / kill} all skin flora (1)	ACCEPT microorganisms / bacteria for skin flora ACCEPT rubbing hands together will {push / damage} microorganisms ACCEPT disinfectant / surfactant ACCEPT higher chance ACCEPT soap cannot destroy viruses	(2)